Introduction

This analysis aims to develop a predictive model for house prices in Ames, Iowa, using an open-source longitudinal Ames Housing dataset, spanning from 2006 to 2010. This dataset consists of 2930 property sales and approximately 80 variables, with SalePrice as the target variables. This reproducible project follows a systematic approach to data exploration, preprocessing, modelling, and evaluation.

The primary objective is to identify key structural and quality-related factors that influence housing prices in Ames and to develop a reliable and interpretable prediction model. Considering the inherent uncertainty and heterogeneity present in real estate markets, the findings of this analysis are intended to provide evidence -based insights rather than definitive price predictions.

Problem Statements

1. What are the key factors influencing house prices in Ames?
2. How strongly do different property features correlate with sale prices?
3. Can we build a reliable model to predict house prices in Ames?
4. Which modelling approach provides the best balance of accuracy and simplicity?

Analysis

Task 2 Descriptive Statistics:

This task provides a statistical overview of the Ames Housing dataset to understand the distribution, scale and variability of numerical variables related to SalePrice. Descriptive statistics serve as a foundation for identifying patterns that justify subsequent regression analysis

```r
#2. EDA: Descriptive Statistics   ----
amesHousing%>%
  select_if(is.numeric)%>% # return numerical variables only
  psych::describe()%>%     # compute descriptive statistics
  knitr::kable(
    align= "c",
    caption= "Descriptive Statistics for Ames Housing",
    format= "html",
    digits= 2
  )%>%
  kableExtra::kable_classic_2(
    #bootstrap_options= c("hover", "stripped", "condensed"),
    html_font= "calibri",
    position= "center",
    font_size= 13  )
```

Descriptive Statistics for Ames Housing

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Order	1	2930	1465.50	845.96	1465.5	1465.50	1086.00	1	2930	2929	0.00	-1.20	15.63
PID	2	2930	714464496.99	188730844.65	535453620.0	712423200.40	12373193.97	526301100	1007100110	480799010	0.06	-1.99	3486655.78
MS.SubClass	3	2930	57.39	42.64	50.0	49.76	44.48	20	190	170	1.36	1.38	0.79
Lot.Frontage	4	2440	69.22	23.37	68.0	68.35	17.79	21	313	292	1.50	11.20	0.47
Lot.Area	5	2930	10147.92	7880.02	9436.5	9481.05	3024.50	1300	215245	213945	12.81	264.39	145.58
Overall.Qual	6	2930	6.09	1.41	6.0	6.08	1.48	1	10	9	0.19	0.05	0.03
Overall.Cond	7	2930	5.56	1.11	5.0	5.47	0.00	1	9	8	0.57	1.48	0.02
Year.Built	8	2930	1971.36	30.25	1973.0	1974.25	37.06	1872	2010	138	-0.60	-0.50	0.56
Year.Remod.Add	9	2930	1984.27	20.86	1993.0	1985.63	20.76	1950	2010	60	-0.45	-1.34	0.39
Mas.Vnr.Area	10	2907	101.90	179.11	0.0	61.14	0.00	0	1600	1600	2.60	9.26	3.32
BsmtFin.SF.1	11	2929	442.63	455.59	370.0	384.08	548.56	0	5644	5644	1.41	6.84	8.42
BsmtFin.SF.2	12	2929	49.72	169.17	0.0	2.04	0.00	0	1526	1526	4.14	18.73	3.13
Bsmt.Unf.SF	13	2929	559.26	439.49	466.0	510.77	415.13	0	2336	2336	0.92	0.40	8.12
Total.Bsmt.SF	14	2929	1051.61	440.62	990.0	1035.05	349.89	0	6110	6110	1.16	9.11	8.14
X1st.Flr.SF	15	2930	1159.56	391.89	1084.0	1127.17	349.89	334	5095	4761	1.47	6.95	7.24
X2nd.Flr.SF	16	2930	335.46	428.40	0.0	272.90	0.00	0	2065	2065	0.87	-0.42	7.91
Low.Qual.Fin.SF	17	2930	4.68	46.31	0.0	0.00	0.00	0	1064	1064	12.11	175.18	0.86
Gr.Liv.Area	18	2930	1499.69	505.51	1442.0	1452.25	461.09	334	5642	5308	1.27	4.12	9.34
Bsmt.Full.Bath	19	2928	0.43	0.52	0.0	0.40	0.00	0	3	3	0.62	-0.75	0.01
Bsmt.Half.Bath	20	2928	0.06	0.25	0.0	0.00	0.00	0	2	2	3.94	14.88	0.00
Full.Bath	21	2930	1.57	0.55	2.0	1.56	0.00	0	4	4	0.17	-0.54	0.01
Half.Bath	22	2930	0.38	0.50	0.0	0.34	0.00	0	2	2	0.70	-1.03	0.01
Bedroom.AbvGr	23	2930	2.85	0.83	3.0	2.83	0.00	0	8	8	0.31	1.88	0.02
Kitchen.AbvGr	24	2930	1.04	0.21	1.0	1.00	0.00	0	3	3	4.31	19.82	0.00
TotRms.AbvGrd	25	2930	6.44	1.57	6.0	6.33	1.48	2	15	13	0.75	1.15	0.03
Fireplaces	26	2930	0.60	0.65	1.0	0.52	1.48	0	4	4	0.74	0.10	0.01
Garage.Yr.Blt	27	2771	1978.13	25.53	1979.0	1980.71	31.13	1895	2207	312	-0.38	1.82	0.48
Garage.Cars	28	2929	1.77	0.76	2.0	1.77	0.00	0	5	5	-0.22	0.24	0.01
Garage.Area	29	2929	472.82	215.05	480.0	468.35	182.36	0	1488	1488	0.24	0.94	3.97
Wood.Deck.SF	30	2930	93.75	126.36	0.0	71.21	0.00	0	1424	1424	1.84	6.73	2.33
Open.Porch.SF	31	2930	47.53	67.48	27.0	33.87	40.03	0	742	742	2.53	10.92	1.25
Enclosed.Porch	32	2930	23.01	64.14	0.0	4.83	0.00	0	1012	1012	4.01	28.42	1.18
X3Ssn.Porch	33	2930	2.59	25.14	0.0	0.00	0.00	0	508	508	11.39	149.63	0.46
Screen.Porch	34	2930	16.00	56.09	0.0	0.00	0.00	0	576	576	3.95	17.81	1.04
Pool.Area	35	2930	2.24	35.60	0.0	0.00	0.00	0	800	800	16.92	299.06	0.66
Misc.Val	36	2930	50.64	566.34	0.0	0.00	0.00	0	17000	17000	21.98	564.85	10.46
Mo.Sold	37	2930	6.22	2.71	6.0	6.16	2.97	1	12	11	0.19	-0.46	0.05
Yr.Sold	38	2930	2007.79	1.32	2008.0	2007.74	1.48	2006	2010	4	0.13	-1.16	0.02
SalePrice	39	2930	180796.06	79886.69	160000.0	170429.15	54856.20	12789	755000	742211	1.74	5.10	1475.84

Observation of Results

The Ames Housing dataset contains 2930 property sales and has 82 variables, with SalePrice as the target variable. There are relatively large ranges for variables such as above-ground living area, total basement area and sale prices as compared to more constrained features like the number of rooms and garage capacity. The distribution of SalePrice is leptokurtic and

positively skewed, which indicates the presence of high-price properties and potential outliers. These values may play a role in modelling, and they should be examined carefully in later stages.

Task 3 Data pre-processing

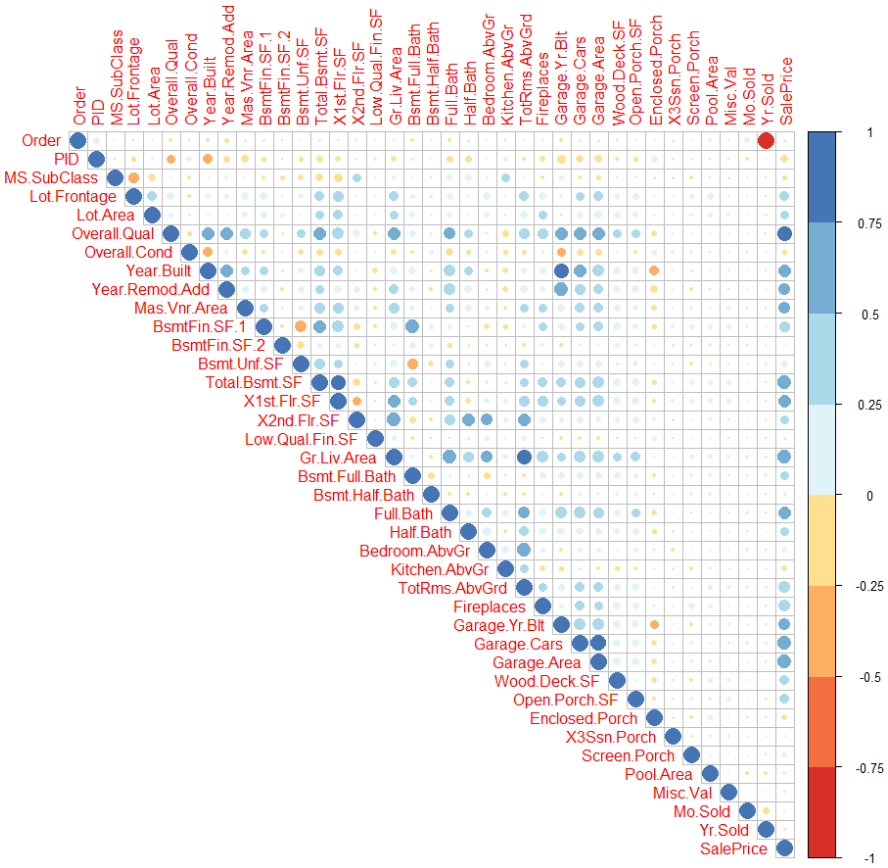
This task focusses on handling missing values through median imputation. The median is used as a better proxy to the mean because several key variables exhibit skewed distributions, making it more robust measure of central tendency. This step ensures that all observations are retained without introducing bias from extreme values.

```
#3. Data cleaning: median imputation for NA values ----
amesHousing <- amesHousing%>%
  mutate(across(where(is.numeric), ~ifelse(is.na(.), median(., na.rm= TRUE), .)))
```

Task 4, 5: Correlation Analysis

These tasks computes and visualise the pairwise correlation matrix for all numerical variables using the cor() function. The correlation matrix highlights the strength and direction of the relationships between predictor variables and SalePrice.

```
#4. Correlation matrix for numeric variables ----
corMatrix<- amesHousing %>%
  select_if(is.numeric)%>%
  cor(use= "pairwise")

##5. Correlation matrix plot ----
#install.packages("RColorBrewer")
#install.packages("corrplot")
library(corrplot)       # plotting the correlation matrix
library(RColorBrewer)   # color palettes
corrplot(corMatrix, type= "upper", col=brewer.pal(n=8, name="RdYlBu"))
```



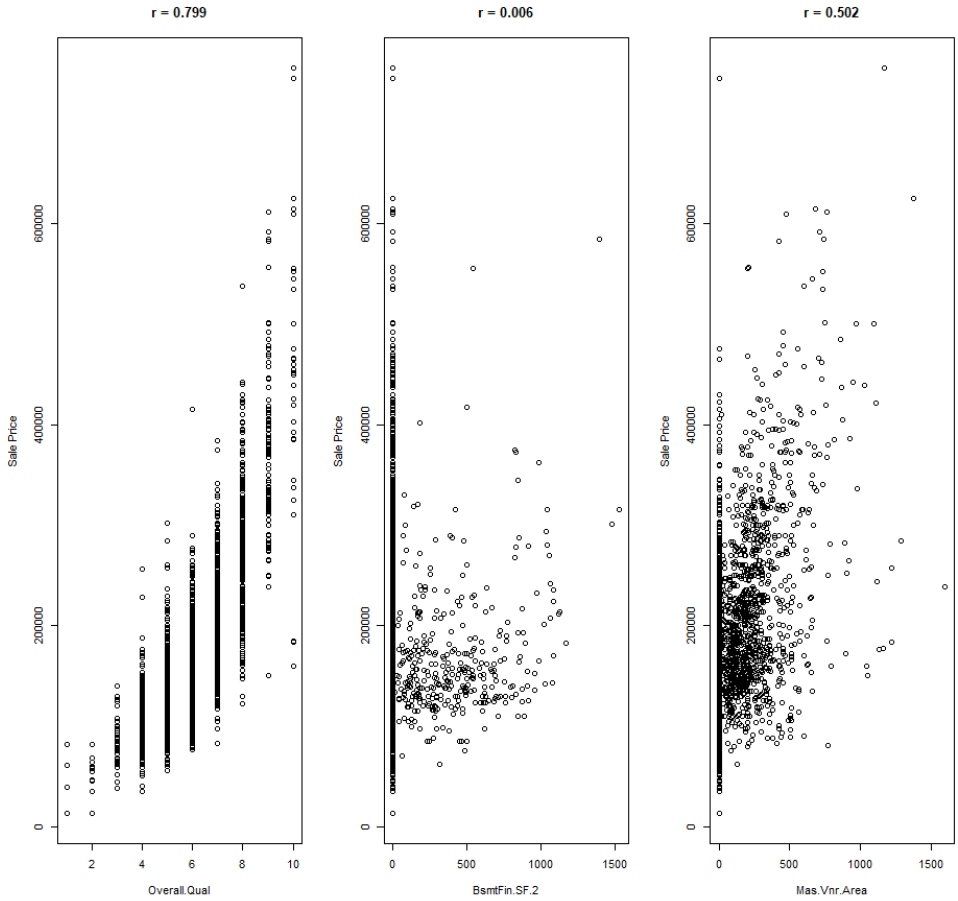
Observation of results

There are strong correlations between SalePrice and variables related to overall quality, living area, garage capacity and basement size. This implies that both property size and perceived quality play a substantial role in determining housing prices. Conversely, several variables exhibit weak correlations which implies limited linear association with sale prices.

Task 6: Scatter plots

This task creates scatter plots between SalePrice and three continuous predictors: The variable with the highest correlation, the variable with the lowest correlation and the variable with a correlation closest to 0.5. These relationships are plotted on the same scale to allow direct comparison for linear patterns.

```r
##6.  Extract correlations with SalePrice ----
saleCor <- corMatrix[, "SalePrice"]                  # get the SalePrice column
saleCor<- saleCor[names(saleCor) != "SalePrice"]   # remove self-correlation

## Find the three variables (highest, lowest, moderate correlation) ----
vars<- list(
  high <- names(which.max(saleCor)),
  low <- names(which.min(abs(saleCor))), # Use abs() to get the coefficient closest to 0
  mid <- names(which.min(abs(saleCor-0.5)))
)

options(scipen= 999) # disabble scientific notation

## Create scatter plots ----
par(mfrow=c(1, 3))    # set plot layout to 1 row, 3 columns
for (i in 1:3){
  varName<- vars[[i]]
  plot(amesHousing[[varName]], amesHousing$SalePrice,
       xlab= varName, ylab= "Sale Price",
       main= paste("r =", round(saleCor[varName], 3))
       )
}
```



Observation of Results

The scatter plots visually back the correlation results. Highly correlated variables show a clear linear trend, while weakly correlated variables show a dispersed pattern. The variable with moderate correlation displays a balances relationship, which supports its inclusion as a potential predictor

Task 7, 8: Regression Analysis

A multiple regression model is fitted with the dependent variable, SalePrice, using the five continuous predictor variables with relatively larger correlation coefficients with the dependent variable. The model estimates the marginal effect of each predictor variable on housing while holding other factors constant

```
> #7. Fit a multiple linear regression model ----
> fit<- lm(SalePrice ~ Overall.Qual + Gr.Liv.Area + Garage.Cars +
+            Garage.Area + Total.Bsmt.SF, data= amesHousing)
> summary(fit)

Call:
lm(formula = SalePrice ~ Overall.Qual + Gr.Liv.Area + Garage.Cars +
    Garage.Area + Total.Bsmt.SF, data = amesHousing)

Residuals:
    Min      1Q  Median      3Q     Max
-512234  -19621   -1231   16636  278498

Coefficients:
                Estimate  Std. Error t value             Pr(>|t|)
(Intercept)   -103655.334    3115.243 -33.274 < 0.0000000000000002 ***
Overall.Qual    24161.818     704.565  34.293 < 0.0000000000000002 ***
Gr.Liv.Area        46.564       1.737  26.808 < 0.0000000000000002 ***
Garage.Cars      9183.681    2076.310   4.423           0.00001009 ***
Garage.Area        33.018       7.267   4.543           0.00000576 ***
Total.Bsmt.SF      33.776       1.971  17.140 < 0.0000000000000002 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 37450 on 2924 degrees of freedom
Multiple R-squared:  0.7806,    Adjusted R-squared:  0.7803
F-statistic:  2081 on 5 and 2924 DF,  p-value: < 0.00000000000000022

>
```

Observation of results

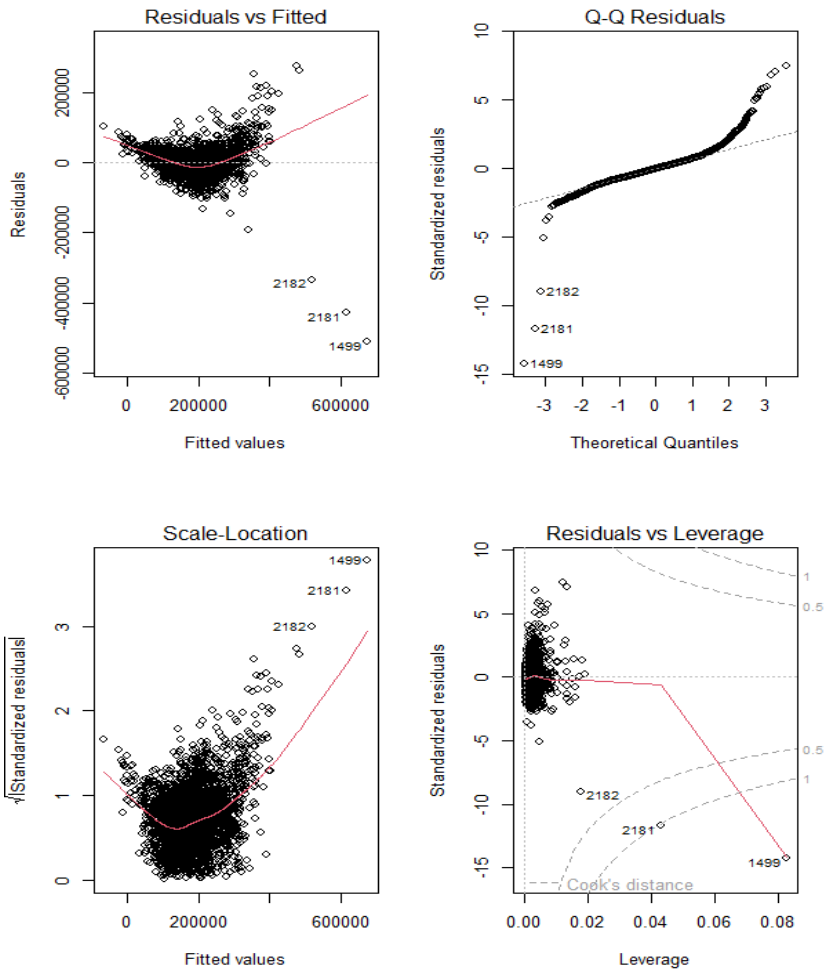
All included predictor variables are statistically significant, indicating that each contributes meaningfully to explain variation in SalePrice within the Ames housing market. The estimated coefficients are rational and aligns with the economic intuition, since improvements in property size, quality and garage capacity are associated with higher sale prices. The regression model is as follows:

SalePrice = -103655.33 + 24161.82 Overall.Qual + 46.56 Gr.Liv.Area + 9183.68 Garage.Cars + 33.02 Garage.Area + 33.78 Total.Bsmt.SF

Task 9: Model diagnostics

This task evaluates key linear model assumptions which include linearity, normality of residuals, independence and constant variance, using diagnostic plots generated by plotting the regression object.

```
#9. Model Diagnostic ----
par(mfrow=c(2,2))  # Set plot layout to 2 row, 2 columns
plot(fit) # regression diagnostics plots
```



Observations of results

Visual inspection of the Residual vs Fitted reviews a moderate positive relationship of the residuals and fitted values thus compromising the linearity assumption. The quantile-quantile plot shows that the predictor variable violets the normality assumption as the

standardised residuals significantly deviates from the diagonal line. The spread of standardised residuals generally increases with the fitted values, indicating varying variances of residuals, thus compromising the homoscedasticity assumption. Lastly, points 2181, 2182, and 1499 can be concluded as influential outliers. These suggests that transformations and further model refinement may be beneficial

Task 10: Multicollinearity

Variable Inflation Factors (VIFs) are computed to investigate dependencies among predictor variables

```
> ##10. Check of multicollinearity  using VIF ----
> #install.packages("car")
> library(car)  # regression diagnostics
> vif(fit)
 Overall.Qual   Gr.Liv.Area   Garage.Cars   Garage.Area Total.Bsmt.SF
     2.064308      1.610248      5.206988      5.099211      1.574145
> |
```

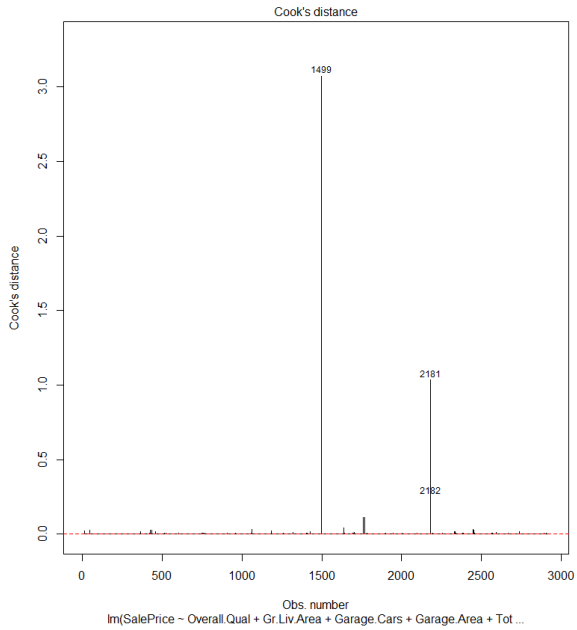
Observation of results

The Garage.Car and Garage.Areas exhibits high multicollinearity. To mitigate this issue and improve model stability and reliability, Garage.Car will be excluded from the model since it has a slightly higher p value than Garage.Area, which is relatively informative in this context

Task 11: Outlier Analysis

This task examines the outliers in the SalePrice and applies high-leverage and influential diagnostics, and Cook's Distance, to determine whether extreme observations disproportionately affect model performance.

```
> ##11. Check for outliers ----
> outlierTest(model= fit)
       rstudent                                    unadjusted p-value                                                  Bonferroni p
1499 -14.806924 0.000000000000000000000000000000000000000000000068814 0.000000000000000000000000000000000000000000020162
2181 -11.997290 0.000000000000000000000000000000000209109999999999980028 0.000000000000000000000000000000000612699999999999998678
2182  -9.116495 0.000000000000000000000140500000000000000179301018476962781 0.000000000000000000000411680000000000000158036084219048 69
1768   7.553516 0.000000000000000563060000000000000261886345942485831983 50 0.000000000016498000000000000000826120422070530935343 4
1761   7.138704 0.000000000000011835999999999999246144688491710894595599 0.000000000346789999999999999875790329673108658425 92
45     6.859749 0.000000000000838580000000000000734301508487078535836190 0.000000000245700000000000000097550439948079770147 1602
1064   5.999596 0.000000000222130000000000017773418217831945753459876869 0.000000650840000000000000231066554778891486421343870
434    5.865816 0.000000000497029999999999968010588879963052022503688 93 0.000014562999999999999312626230985046049681841396
1638   5.790904 0.000000000774560000000000000218310092225948437771876342 6 0.000022694999999999999870301664595118040779198054 2
433    5.543879 0.000000000322200000000000000226323126906180505102383904 16 0.000094408999999999999968278985296166183616151101 89
> |
```



Observation of results

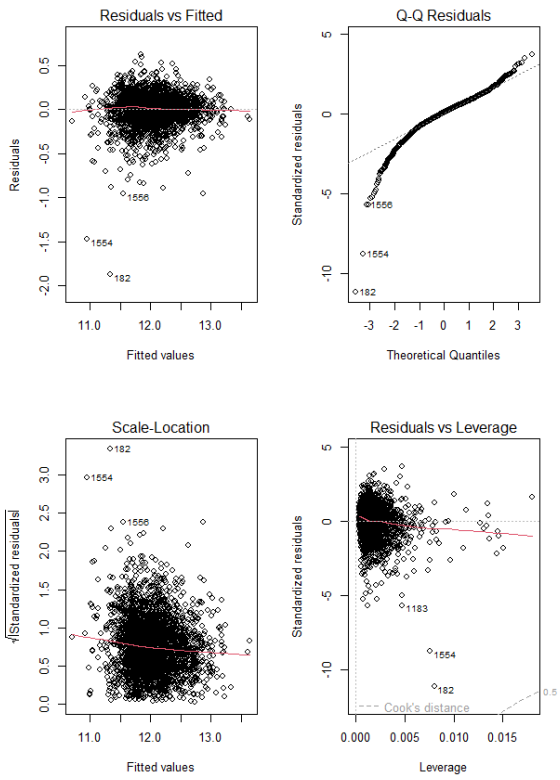
Several outliers are identified, with three influential points exerting excessive influence on the regression model. These points are flagged for removal to prevent distortion of coefficient estimates

Task 12 Model Improvement:

This task removes three influential outliers, drops the Garage.Areas to cater for multicollinearity and applies logarithmic transformation to improve normality and linearity. The Akaike Information Criterion (AIC) is used to compare the refined model against the original model

```
> ##12. Remove influential outliers identified via Cook's Distance ----
> outliers <- c(1499, 2181, 2182)     # define influential outliers
> amesHousing <- amesHousing[-outliers, ]  # remove outliers to reduce undue leverage on the model
>
> ## Transformation check ----
> summary(powerTransform(amesHousing$SalePrice))
bcPower Transformation to Normality
                    Est Power Rounded Pwr Wald Lwr Bnd Wald Upr Bnd
amesHousing$SalePrice    0.0074           0      -0.0504       0.0652

Likelihood ratio test that transformation parameter is equal to 0
 (log transformation)
                          LRT df    pval
LR test, lambda = (0) 0.06363572  1 0.80084

Likelihood ratio test that no transformation is needed
                          LRT df                pval
LR test, lambda = (1) 965.7734  1 < 0.000000000000000222
>
> ## Apply log normal transformation to improve normality ----
> amesHousing$LogSalePrice <- log(amesHousing$SalePrice)
>
> ## Fit the cleaned model with Garage.Car removed due to high VIF ----
> fitLog<- lm(LogSalePrice ~ Overall.Qual + Gr.Liv.Area + Garage.Area
+            + Total.Bsmt.SF, data= amesHousing)
>
> ## Review model summary to assess coefficient stability and fit ----
> summary(fitLog)

Call:
lm(formula = LogSalePrice ~ Overall.Qual + Gr.Liv.Area + Garage.Area +
    Total.Bsmt.SF, data = amesHousing)

Residuals:
     Min       1Q   Median       3Q      Max
-1.87681 -0.08038  0.01696  0.10279  0.62792

Coefficients:
                Estimate   Std. Error t value             Pr(>|t|)
(Intercept)   10.493137142  0.014135043  742.35 <0.0000000000000002 ***
Overall.Qual   0.133825323  0.003115305   42.96 <0.0000000000000002 ***
Gr.Liv.Area    0.000243689  0.000007962   30.61 <0.0000000000000002 ***
Garage.Area    0.000304236  0.000018588   16.37 <0.0000000000000002 ***
Total.Bsmt.SF  0.000195150  0.000009159   21.31 <0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.169 on 2922 degrees of freedom
Multiple R-squared:  0.8285,    Adjusted R-squared:  0.8282
F-statistic:  3529 on 4 and 2922 DF,  p-value: < 0.00000000000000022

>
> ## Recheck multicollonearity ----
> vif(fitLog)
 Overall.Qual    Gr.Liv.Area   Garage.Area Total.Bsmt.SF
     1.966078       1.572063       1.619530       1.533508
```

Observation of results



The refined model accounts for 82.82% of the variability in sale price, which is better than the original models 78.03% coefficient of multiple determination. This indicates an improved

balance between fit and simplicity relative to the original model specification. The lower AIC value for the refined model (-2094.162) also suggests that its relatively better in explaining the SalePrice data than the original proposed model with 70032.926

AIC coefficient.

Task 13: Stepwise Regression

The stepwise regression using a bidirectional search is applied to identify the optimal subset of predictor variables that best explains SalePrice while balancing simplicity and explanatory power

```r
#13. Stepwise regression ----
library(MASS) # stats functions
bestModel<- stepAIC(lm(SalePrice ~ ., data= amesHousing[, sapply(amesHousing, is.numeric)]),
                    direction= "both")
```

```
Step:  AIC=-12000.55
LogSalePrice ~ Order + MS.SubClass + Lot.Frontage + Lot.Area +
    Overall.Qual + Overall.Cond + Year.Built + Year.Remod.Add +
    BsmtFin.SF.1 + BsmtFin.SF.2 + Bsmt.Unf.SF + Total.Bsmt.SF +
    X1st.Flr.SF + X2nd.Flr.SF + Low.Qual.Fin.SF + Bsmt.Full.Bath +
    Full.Bath + Bedroom.AbvGr + Kitchen.AbvGr + TotRms.AbvGrd +
    Fireplaces + Garage.Yr.Blt + Garage.Cars + Garage.Area +
    Wood.Deck.SF + Enclosed.Porch + Screen.Porch

                 Df Sum of Sq    RSS    AIC
<none>                       47.591 -12001
- Full.Bath       1    0.0420 47.633 -12000
+ Pool.Area       1    0.0185 47.573 -12000
+ PID             1    0.0168 47.575 -12000
+ Open.Porch.SF   1    0.0146 47.577 -12000
- Wood.Deck.SF    1    0.0528 47.644 -11999
+ X3Ssn.Porch     1    0.0099 47.582 -11999
+ Half.Bath       1    0.0059 47.585 -11999
+ Bsmt.Half.Bath  1    0.0058 47.586 -11999
+ Yr.Sold         1    0.0029 47.589 -11999
+ Mo.Sold         1    0.0011 47.590 -11999
+ Mas.Vnr.Area    1    0.0000 47.591 -11999
+ Misc.Val        1    0.0000 47.591 -11999
- Total.Bsmt.SF   1    0.0719 47.663 -11998
- Bsmt.Unf.SF     1    0.0815 47.673 -11998
- Garage.Area     1    0.0828 47.674 -11998
- Garage.Yr.Blt   1    0.0835 47.675 -11997
- BsmtFin.SF.2    1    0.0843 47.676 -11997
- BsmtFin.SF.1    1    0.0885 47.680 -11997
- Low.Qual.Fin.SF 1    0.1070 47.698 -11996
- TotRms.AbvGrd   1    0.1087 47.700 -11996
- Order           1    0.2021 47.793 -11990
- Enclosed.Porch  1    0.2270 47.818 -11989
- Bsmt.Full.Bath  1    0.2301 47.821 -11988
- Bedroom.AbvGr   1    0.2346 47.826 -11988
- Garage.Cars     1    0.2421 47.834 -11988
- Lot.Frontage    1    0.3319 47.923 -11982
- MS.SubClass     1    0.3911 47.982 -11979
- Kitchen.AbvGr   1    0.4828 48.074 -11973
- Screen.Porch    1    0.5143 48.106 -11971
- Year.Remod.Add  1    0.5338 48.125 -11970
- Fireplaces      1    0.8010 48.392 -11954
- Lot.Area        1    0.9027 48.494 -11948
- Year.Built      1    5.7649 53.356 -11668
- X1st.Flr.SF     1    6.0153 53.607 -11654
- Overall.Cond    1    6.9027 54.494 -11606
- X2nd.Flr.SF     1    9.2750 56.866 -11481
- Overall.Qual    1   12.3694 59.961 -11326
>
```

Observation of results

The bidirectional stepwise procedure selects a reduced set of predictors that maintains a strong explanatory performance while improving interpretability. The regression model is as follows:

LogSalePrice =

1.72 + 0.00000988 Order -0.000347 MS SubClass + 0.000639 Lot Frontage + 0.00000254 Lot Area + 0.082 Overall Qual + 0.054 Overall Cond + 0.00307 Year Built + 0.00101 Year Remod Add + 0.00194 BsmtFin SF 1 + 0.00189 BsmtFin SF 2 + 0.00186 Bsmt Unf SF -0.00175 Total Bsmt SF + 0.000274 X1st Flr SF + 0.000274 X2nd Flr SF + 0.000136 Low Qual Fin SF + 0.0237 Bsmt Full Bath + 0.0107 Full Bath -0.0165 Bedroom AbvGr -0.0747 Kitchen AbvGr + 0.00816 TotRms AbvGrd + 0.0319 Fireplaces + 0.000388 Garage Yr Blt + 0.0289 Garage Cars + 0.0000587 Garage Area + 0.00003696 Wood Deck SF + 0.0001518 Enclosed Porch + 0.000246 Screen Porch

Task 14: Model comparison

This tsk compares our proposed, refined model with the stepwise model on predictive accuracy, simplicity, and interpretability to determine the preferred final model

```
> #14. Compare models ----
> AIC(bestModel, fitLog)
          df       AIC
bestModel 29 -3692.085
fitLog     6 -2094.162
> ## Multicollinearity comparison ----
> vif(bestModel)
          Order      MS.SubClass     Lot.Frontage         Lot.Area     Overall.Qual     Overall.Cond       Year.Built  Year.Remod.Add     BsmtFin.SF.1     BsmtFin.SF.2
       1.012915         1.637009         1.558979         1.263211         3.144530         1.527876         4.365197         2.409621    23970.838859     3556.903041
    Bsmt.Unf.SF   Total.Bsmt.SF      X1st.Flr.SF      X2nd.Flr.SF Low.Qual.Fin.SF  Bsmt.Full.Bath        Full.Bath    Bedroom.AbvGr   Kitchen.AbvGr   TotRms.AbvGrd
   23991.045048    22151.722154         5.221174         4.331199         1.078827         1.956935         2.447519         2.341500         1.551340         4.357836
      Fireplaces   Garage.Yr.Blt      Garage.Cars      Garage.Area     Wood.Deck.SF  Enclosed.Porch     Screen.Porch
       1.554648         3.251452         5.848738         5.578002         1.203413         1.222944         1.083294
> vif(fitLog)
 Overall.Qual   Gr.Liv.Area   Garage.Area Total.Bsmt.SF
     1.966078      1.572063      1.619530      1.533508
> |
```

Observation of results

While both models perform well, the stepwise model produces a more parsimonious model specification with comparable explanatory performance, making it preferable for practical application. This is evidenced by a lower AIC value of -3692 for the stepwise model as compared to a slightly larger AIC value for our proposed model of -2094. A lot of multicollinearities also exist in the predictor variables for the stepwise model, which might make it complicated to figure out which factors drive the sale price without confounding interdependences between variables

Conclusion

This analysis demonstrates that housing prices in Ames, Iowa, are fundamentally driven by property size, quality, and garage capacity. Using exploratory analysis, regression modelling and diagnostic evaluation, a relatively reliable predictive model was developed. Handling multicollinearity, outliers, and distributional issues significantly improved model performance

Although no single model can perfectly match the complexity of real estate markets, our proposed model performed well as compared to the superior stepwise regression model in providing a robust and evidence-based tool for explaining the price dynamics. Future work could extend this analysis by incorporating non-linear methods to enhance predictive accuracy as most linear assumptions were compromised during the analysis.